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Rethinking Token Reduction for Large Vision-Language Models

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Abstract

Large Vision-Language Models (LVLMs) excel in visual understanding and reasoning, but the excessive visual tokens lead to high inference costs. Although recent token reduction methods mitigate this issue, they mainly target single-turn Visual Question Answering (VQA), leaving the more practical multi-turn VQA (MT-VQA) scenario largely unexplored. MT-VQA introduces additional challenges, as subsequent questions are unknown beforehand and may refer to arbitrary image regions, making existing reduction strategies ineffective. Specifically, current approaches fall into two categories: prompt-dependent methods, which bias toward the initial text prompt and discard information useful for subsequent turns; prompt-agnostic ones, which, though technically applicable to multi-turn settings, rely on heuristic reduction metrics such as attention scores, leading to suboptimal performance. In this paper, we propose a learning-based prompt-agnostic method, termed MetaCompress, overcoming the limitations of heuristic designs. We begin by formulating token reduction as a learnable compression mapping, unifying existing formats such as pruning and merging into a single learning objective. Upon this formulation, we introduce a data-efficient training paradigm capable of learning optimal compression mappings with limited computational costs. Extensive experiments on MT-VQA benchmarks and across multiple LVLM architectures demonstrate that MetaCompress achieves superior efficiency–accuracy trade-offs while maintaining strong generalization across dialogue turns. Our code is available at https://github.com/MArSha1147/MetaCompress.

1. Introduction

LVLMs [1, 4, 35–37, 69, 70] have become powerful AI systems enabling natural human interaction with visual data such as images and videos. They encode both textual and visual modalities into tokens jointly processed by a unified Large Language Model (LLM). Recent works [37, 70] further extend LVLMs toward multi-scale visual inputs that integrate both global and local tokens to enhance visual understanding. However, visual tokens greatly increase computation and memory costs, as token numbers grow by thousands and attention scales quadratically with sequence length [13, 30], making low-latency or resource-constrained deployment challenging [64].

Although numerous token reduction techniques [9, 49, 65] have been proposed and have achieved considerable success, they are primarily developed for single-turn VQA. Meanwhile, the more realistic MT-VQA setting, which involves multi-round conversational question answering, remains largely underexplored thus far. Compared to single-turn VQA, which focuses on answering one-shot questions and can greedily discard image tokens irrelevant to the current query, MT-VQA poses additional challenges due to its open-ended nature. In MT-VQA, future questions are entirely unpredictable, and their relevant regions may arise anywhere in the image, making existing token reduction methods difficult to apply directly.

Existing token reduction methods can be broadly categorized into prompt-dependent and prompt-agnostic approaches. Prompt-dependent methods, such as FastV [9], retain tokens that are highly relevant to only the first-turn question prompt. This strategy may inadvertently discard visual information that could be crucial for answering subsequent questions; for example, the first question might focus on the foreground while later questions may reference the background. In contrast, prompt-agnostic approaches like PruMerge [49] reduce tokens solely based on attention scores within the image sequence itself, which are technically applicable to multi-turn interactions. However, a critical limitation of existing prompt-agnostic methods lies in their reliance on heuristic reduction criteria derived from human priors, and the lack of theoretical guidance, often resulting in suboptimal performance.

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In response to this, we propose a learning-based prompt-agnostic token reduction approach, termed MetaCompress, which overcomes the drawbacks of heuristic designs. To achieve this, a key question is how the learning objective should be defined. By analyzing the reduction formats of current approaches, including both pruning and merging, we find that they can be unified by formulating the visual token reduction task as an optimization problem. The goal is to identify an optimal compression mapping of the input visual tokens, under conditions such as language conditioning in prompt-dependent approaches and image-only conditioning in prompt-agnostic approaches, so that the model's responses exhibit minimal discrepancy after token reduction.

Based on this formulation, we first simplify the problem by learning an optimal compression matrix for each image and conduct a preliminary investigation into the guiding role of attention information, as commonly employed in previous methods. Surprisingly, our findings reveal that the tokens retained by the learned matrix do not exhibit an obvious relationship with the heuristic attention cues commonly used in prior methods, such as [CLS] token attention and prompt-token attention, further validating that heuristic reduction criteria are suboptimal.

Furthermore, to fully implement MetaCompress, a practical challenge arises from the need to generate multiple compression matrices, since actual image inputs can vary in resolution. And learning specific compression matrices for every possible resolution is not an especially elegant or practical solution. To address this, we ultimately design to learn a compression matrix generator compatible with dynamic resolutions, trained in a data-efficient paradigm. Extensive experiments on three MT-VQA benchmarks using five LVLM architectures demonstrate that MetaCompress outperforms existing token reduction methods while achieving high computational efficiency.

The contributions of our paper are summarized as follows:

- We first explore token reduction in the MT-VQA scenario, revealing that heuristic methods relying on visual token attention scores are suboptimal.
- We propose MetaCompress, a novel learning-based and prompt-agnostic token reduction method, overcoming the reliance on suboptimal heuristic reduction criteria.
- MetaCompress leverages a data-efficient training paradigm to learn the optimal compression mapping for visual sequences, demonstrating the effectiveness and efficiency through extensive experiments.

2. Related Work

2.1. Efficient Large Vision-Language Models

LVLMs. Transformers [57] have unified architectures across language [6, 16, 46] and vision [7, 17, 23, 26, 53], then CLIP [45] bridges both modalities through contrastive pre-training, enabling zero-shot visual understanding. Based on this, LVLMs [2, 35–37, 58, 66, 70] integrate visual encoders with large language models to perform multimodal tasks such as captioning and VQA. LLaVA [35, 36] achieves image-to-text generation by feeding CLIP-encoded visual tokens and language tokens into an LLM *e.g.*, Llama [54, 55], but its fixed global resolution restricts fine-grained perception. Recent models such as LLaVA-NeXT [37] and InternLM-XComposer-2.5 [70] enhance visual understanding by incorporating multi-scale visual inputs that combine global and local tokens, but this substantially increases token counts, leading to heavy memory and computation overhead in multi-head attention and auto-regressive decoding, particularly on resource-constrained devices.

Model Quantization. To deploy LVLMs to low-memory devices such as mobile while preserving the model's performance, a natural approach is to quantize the model and inference process into 4/8-bit [15, 19, 67] or even 1-bit [42]. Another line of work focuses on reducing the computational burden of MHA by employing efficient attention mechanisms [10, 12, 13, 32] or sparse attention [11, 59, 60]. However, quantization methods are limited by optional fine-tuning and hardware support, and more importantly, they do not solve the overall computational inefficiency caused by the increasing number of visual tokens.

Model Pruning. Model pruning [20, 40, 43, 71] and knowledge distillation [25, 48, 61] methods compress the given model to arbitrary size by removing redundant parameters or transferring knowledge from a large model to a smaller one. These approaches are generally effective at reducing model size and inference cost, but often require careful hyperparameter tuning and expensive retraining procedure.

2.2. Visual Token Reduction

Recent studies show that image representations contain substantial redundancy [14, 24], enabling feature reduction without significant performance loss [5, 8, 29]. This observation has motivated the development of token reduction techniques for LVLMs, which can be broadly categorized into prompt-dependent and prompt-agnostic approaches. Prompt-dependent methods, such as FastV [9], identify redundant tokens by measuring their attention to language prompts and remove them at specific layers. FitPrune [65] identifies redundant tokens by minimizing the divergence of attention distributions before and after pruning. IVTP [27] and TRIM [51] employ CLIP's text encoder to guide token reduction. However, these methods are less applicable to general scenarios such as MT-VQA, as they require re-compression for each question. In contrast, prompt-agnostic methods rely solely on image sequences and are technically applicable to MT-VQA tasks. Nevertheless, existing approaches like LLaVA-PruMerge [49] overlook compatibility with modern LVLMs that incorporate multi-scale vision towers (*e.g.*,

LLaVA-NeXT). More importantly, these methods rely heavily on heuristic reduction criteria derived from human priors, which often lead to suboptimal performance. To address this, this paper introduces *a novel learning-based, prompt-agnostic token reduction method that avoids heuristic designs* (*e.g.*, attention to [CLS] or other tokens as reduction guidance, which will be shown suboptimal later in this paper) and can integrate seamlessly with modern LVLMs.

3. Preliminaries

In this section, we first give a brief review of the inference process of LVLMs, particularly in the context of multi-turn dialogue scenarios. We then introduce the problem definition of visual sequence compression mapping.

3.1. Large Vision-Language Models

Given an input image I_{IMG}, LVLMs are required to generate a series of responses $(R_1, \dots, R_t)$ to user's prompts $(P_1, \dots, P_t)$. The image and the language context are tokenized separately by a vision tower $T_{\text{IMG}}(\cdot)$, *e.g.*, vision Transformer (ViT) [17, 45] and a language tokenizer $T_{\text{TXT}}(\cdot)$, *e.g.*, SentencePiece [31]. The tokenized image and language sequence are then embedded into a common space by a vision projector $V_{\text{IMG}}(\cdot)$ and an embedding layer $E_{\text{TXT}}(\cdot)$, producing X_{IMG} and X_{TXT}, respectively.

To fully capture detailed information, current prevalent LVLMs, such as LLaVA-NeXT and InternLM-XComposer-2.5, employ a ViT to encode images from both global and local views, generating multi-scale visual sequences. Despite enhancing the model's capability to capture the details of the image, such an approach significant increase the token number, severely impairing the inference efficiency due to the $O(n^2)$ computational and memory complexity of MHA [57].

To increase LLMs' inference efficiency, KV cache methods [34, 62] are proposed for reusing intermediate attentions in the auto-regressive decoder. Specifically, for generating the i-th response token with query q_i, the original computation is to concatenate the previous queries for MHA layer

$$\text{MHA}(Q_i, K_i, V_i) = \sigma\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i, \tag{1}$$

where $\{Q, K, V\}_i = (\{Q, K, V\}_{i-1} | \{q, k, v\}_i)$ are the concatenated inputs, and σ denotes the row-wise Softmax operation. To decrease the computation complexity, KV caches store the intermediate key-value pairs $\{K, V\}_{i-1}$ and only compute the attention for q_i:

$$\text{MHA}(q_i, k_i, v_i) = \sigma\left(\frac{q_i (K_{i-1}|k_n)^\top}{\sqrt{d_k}}\right)(V_{i-1}|v_i), \tag{2}$$

which significantly reduce the computation burden. Such techniques can be seamlessly integrated with the MT-VQA setting, where the caches are reused across multiple turns.

3.2. Visual Token Reduction

However, the aforementioned cache mechanism is still insufficient to address the memory and computation overhead caused by the large number of image tokens, resulting in an $O(n^2)$ cost for generating the first token and an $O(nT)$ cost for producing T tokens during multi-turn dialogues.

To alleviate the issue, token reduction methods are proposed to compress the image sequence. For simplicity, we only consider reducing image tokens right before feeding into the LLM, *e.g.*, LLama [54], which can be formulated as

$$\tilde{X}_{\text{IMG}} = \mathcal{P}_{\text{reduce}}(X_{\text{IMG}} | I_{\text{IMG}}, I_{\text{TXT}}), \tag{3}$$

where guiding information is extracted from the input image I_{IMG} and the language context I_{TXT}. Depending on whether to rely on the prompt I_{TXT}, token reduction methods can be categorized into prompt-dependent and prompt-agnostic methods. However, in real-world applications, LVLMs are often required to respond to multiple prompts. Prompt-dependent methods, however, tend to bias toward the initial query and discard information beneficial for subsequent turns, leading to suboptimal performance in multi-turn dialogue scenarios. Furthermore, many existing methods requires the intermediate attention matrices in MHA layers to guide token reduction, whereas modern LVLMs commonly employ FlashAttention [12, 13] or Memory-Efficient Attention [32], which do not support returning them.

To further investigate the optimal token reduction strategy, we first unify the token pruning and merging methods by formulating the reduction process as a linear projection to the input X_{IMG}:

$$\tilde{X}_{\text{IMG}} = P X_{\text{IMG}}, \tag{4}$$

where $P \in \mathbb{R}_+^{m \times n}$ ($m \ll n$) is the sparse compression matrix. In Section 4, we set P as a learnable matrix for each image. By optimizing P in a data-driven manner, we first explore the relationship between the retained tokens and the attention weights employed by heuristics-designed methods. Then in Section 5, we present a novel token reduction method that does not rely on the intermediate attention matrices, while can be seamlessly integrated with modern LVLMs.

4. Which Tokens to Keep?

To objectively analyze the optimal token reduction scheme without relying on hand-crafted designs, we start by looking at a simpler case: *Given an input image I_{IMG} and a conversation context I_{TXT}, find the optimal compression matrix P^* as defined in Equation (4) so that the response discrepancy between the LLM using the compressed and uncompressed visual sequence is minimized.*

To achieves this, let $P_{\text{raw}} \in \mathbb{R}^{m \times n}$ be the trainable reduction parameters, with each element independently drawn from Gaussian distribution $\mathcal{N}(0, \sigma_{\text{raw}}^2)$. We normalize P_{raw}

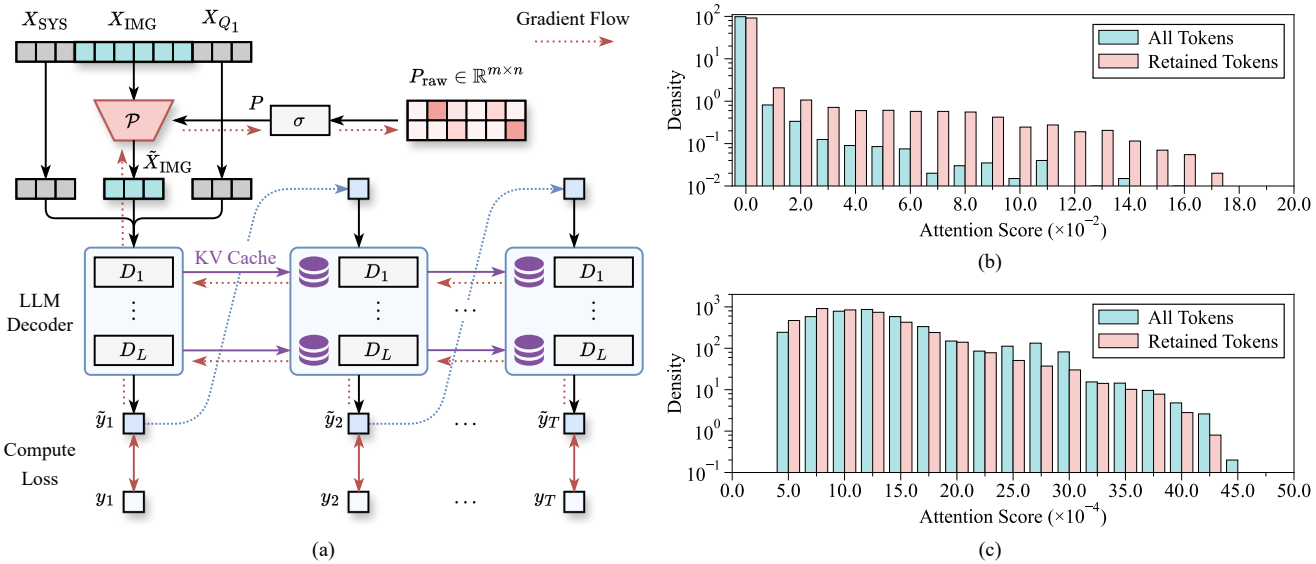


Figure 1. (a) Overall pipeline of the compression projection training process. (b) Attention distribution over the [CLS] token for retained and all visual tokens. The image tokens are extracted from the last layer of the vision tower of LLaVA-1.5-13b running on VQA-v2 dataset. (c) Attention distribution over the prompt tokens for retained and all visual tokens. The attention scores are averaged to prompt tokens extracted from the first layer of the LLM decoder.

with row-wise Softmax to obtain the compression matrix $P = \sigma(P_{\text{raw}})$. Let $p(y|X_{\text{IMG}}, X_{\text{TXT}})$ denotes the original prediction distribution T tokens $y = (y_1, \ldots, y_T)$. Then we force the LLM to generate T tokens $p(\tilde{y}|\tilde{X}_{\text{IMG}}, X_{\text{TXT}})$ using the compressed visual sequence $\tilde{X}_{\text{IMG}}$. Figure 1a illustrates the overall training pipeline, where P_{raw} is trained to minimize the KL divergence between the two response distributions $\mathcal{L}_{\text{pred}} = D_{\text{KL}}(p(y)\|p(\tilde{y}))$ and the distribution entropy $\mathcal{L}_{\text{entropy}} = \frac{1}{m}\sum_{i=0}^{m} \mathcal{H}(P_{i,:})$.

The training objective is formulated as

$$P^* = \arg\min_{P_{\text{raw}}} \mathcal{L}_{\text{pred}} + \alpha \mathcal{L}_{\text{entropy}}. \tag{5}$$

The training algorithm and detailed implementation are provided in Section 8.

Figure 1b and 1c visualize the attention distribution over the [CLS] and prompt tokens, respectively. Despite that some tokens with high attention to the [CLS] token are retained (accounting for approximately 1.71% of the total retained tokens), *the vast majority of the retained tokens are unrelated to their attention scores*, especially with regards to the attention to the language prompts. More results are in Section 6.6, drawing to the same conclusion. This observation suggests that using attention scores as guidance for token reduction is suboptimal in the MT-VQA scenario, which explains the experimental results in Section 6.2, where token pruning methods such as FastV perform worse than uniform or even random pruning. Therefore, it is essential to explore a novel token reduction approach that does not rely on heuristic metrics such as attention scores, while being seamlessly compatible with modern LVLMs.

5. Method

Results from Section 4 inspire us to construct the compression matrix in a data-driven manner. To this end, we propose MetaCompress, a lightweight module learning the compression matrix P conditioned only on the input image for MT-VQA scenarios. Section 5.1 details the MetaCompress module, Section 5.2 provides theoretical analysis, and Section 5.3 presents the optimization objective and training algorithm.

5.1. MetaCompress

Our goal is to learn a compression matrix generator $\mathcal{P}_{\text{meta}}$ in a data-driven manner, so that the overall prediction discrepancy on the given dataset $\mathcal{D} = \{(I^{(i)}_{\text{IMG}}, I^{(i)}_{\text{TXT}})\}_{i=1}^{N}$ is minimized. To this end, we propose a lightweight meta generator $\mathcal{P}_{\text{meta}}$ which computes a compression matrix $P = \mathcal{P}_{\text{meta}}(X_{\text{IMG}})$ for each input image I_{IMG}, independent of the prompt. One major challenge is that $\mathcal{P}_{\text{meta}}$ is required to generate the compression matrix P, whose shape can adapt to the varying length of X_{IMG}, thereby accommodating multiple resolution scales for LVLMs such as LLaVA-NeXT and InternLM-XComposer-2.5.

Figure 2 shows the overall architecture of $\mathcal{P}_{\text{meta}}$, which consists of a position embedding layer, a query down-sample projection $\tilde{D}_q$, a key projection D_k, and a weighted inner product layer. The core idea is to compute the inner product between the spatially down-sampled queries $\tilde{X}_q \in \mathbb{R}^{m \times d_c}$ and keys $X_k \in \mathbb{R}^{n \times d_c}$ to get the compression matrix $P \in \mathbb{R}^{m \times n}$. Here, queries

$$\tilde{X}_q = \tilde{D}_q(X_{\text{IMG}} + E_{\text{pos}}) = \text{Pool}(X_{\text{IMG}} + E_{\text{pos}}|k, s)W_q \tag{6}$$

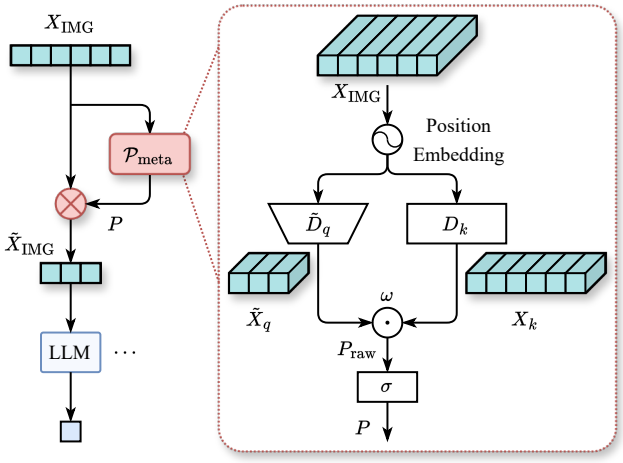


Figure 2. Illustration of our proposed MetaCompress, where module $\mathcal{P}_{\text{meta}}$ generate the compression projection P solely according to the image sequence X_{IMG}.

are down-sampled from the image sequence encoded with absolute position embeddings E_{pos} by average pooling $\text{Pool}(\cdot|k, s)$ with kernel size k and stride s[1], and keys

$$X_k = D_k(X_{\text{IMG}} + E_{\text{pos}}) = (X_{\text{IMG}} + E_{\text{pos}})W_k \quad (7)$$

are linearly projected from X_{IMG} for computational efficiency (by setting $d_c \ll d$). Finally, the computation of compression matrix P is formulated as:

$$P = \sigma\left(\frac{\tilde{X}_q \,\text{diag}(\omega) X_k^\top}{\sqrt{d_c}}\right), \quad (8)$$

where diagonal matrix $\omega \in \mathbb{R}^{d_c}$ is learnable.

Regarding the module placement, following the setting of LLaVA-PruMerge [49], we apply our reduction module only before the LLM decoder, although MetaCompress can in principle be inserted at any layer. This placement reduces the additional MHA computation incurred in earlier LLM layers compared with inserting it at intermediate layers, which is particularly beneficial for long visual inputs such as videos. Moreover, our work focuses on developing a learning-based reduction method rather than a full-stage compression strategy across both the vision tower and the LLM, as explored in IVTP [27]. Such full-stage optimization requires costly pretraining and instruction-tuning, which we leave as future work under our lightweight framework.

For the module design, since our primary goal is to reduce the inference burden of LVLMs, we intentionally avoid constructing complex reduction modules, such as those required for auto-regressive generation, as they would significantly increase latency and reduce computational efficiency. As

[1]Section 8.2 provides the detailed implementation for down-sampling X_{IMG} to arbitrary length m.

Algorithm 1 Training algorithm for MetaCompress.

Require: $E_{\text{TXT}}(\cdot)$: the language encoder; $V_{\text{IMG}}(\cdot)$: the image encoder; $\text{LLM}(\cdot,\cdot)$: the vision-language decoder; $\mathcal{P}_{\text{meta}}(\cdot|\Theta)$: the proposed MetaCompress module with learnable parameters Θ.

for $(I_{\text{IMG}}, I_{\text{TXT}}) \in \mathcal{D}_{\text{train}}$ **do**
  $X_{\text{TXT}} \leftarrow E_{\text{TXT}}(I_{\text{TXT}})$
  $X_{\text{IMG}} \leftarrow V_{\text{IMG}}(I_{\text{IMG}})$
  $\tilde{X}_{\text{IMG}} \leftarrow \mathcal{P}_{\text{meta}}(X_{\text{IMG}}|\Theta)$ # with gradients
  $\boldsymbol{y} \leftarrow \text{LLM}(X_{\text{TXT}}, X_{\text{IMG}})$
  $\tilde{\boldsymbol{y}} \leftarrow \text{LLM}(X_{\text{TXT}}, \tilde{X}_{\text{IMG}})$ # with gradients
  Compute the final loss and gradient ∇_Θ w.r.t. Θ.
  Update Θ with SGD optimizer.
end for

the first learning-based token reduction framework, there are currently few non-learning approaches available for direct comparison. Nevertheless, we further discuss the relationship between our method and other data-driven approaches for efficient model inference in Section 12.

5.2. Theoretical Analysis

Now we provide a theoretical analysis of MetaCompress to explain the design motivation and further introduce the optimization objectives and constraints. To begin with, we expand Equation (8) as

$$P_{\text{raw}} = \text{Pool}(X|k, s)W_q \,\text{diag}(\omega) W_k^\top X^\top. \quad (9)$$

Further, suppose all elements in W are drawn independently from a Gaussian distribution $\mathcal{N}(0, \sigma_c^2)$ and with a specific initialization (*i.e.*, $W_q = W_k$), MetaCompress will initially behave as a weighted pooling of the input image sequence (controlled by the kernel size k), as we prove in Section 9. Moreover, the meta generator will learn, in a data-driven manner, how to select and merge the visual tokens to minimize the prediction discrepancy. Since we do not rely on any annotation for the compression matrix, the low-rank positive semi-definite form presented in Equation (9) provides a good starting point and facilities gradient decent optimization.

5.3. Training MetaCompress

Similar to the training objective introduced in Section 4, we train MetaCompress by minimizing the prediction discrepancy $\mathcal{L}_{\text{pred}}$ with additional sparsity regularization $\mathcal{L}_{\text{entropy}}$. However, due to the lack of ground-truth compression matrix, the generated compression matrix P tends to collapse to trivial solutions where the compressed tokens all derive from the same input source. To avoid this, we add a collapse regularization term $\mathcal{L}_{\text{collapse}} = \max_j \sum_{i=1}^{m} P_{i,j}$. Therefore, the final optimization objective is

$$\mathcal{L} = \mathcal{L}_{pred} + \alpha_{\text{entropy}}\mathcal{L}_{\text{entropy}} + \alpha_{\text{collapse}}\mathcal{L}_{\text{collapse}}, \quad (10)$$

where α_{entropy} and α_{collapse} are hyperparameters. Algorithm 1 delineates the training procedure.

Table 1. The comparison of visual token reduction methods on three MT-VQA benchmarks with the reduction rate of 90%. The best and the second-best results are highlighted in bold and underline, respectively.

Model	Method	MT-VQA-v2				MT-GQA				ConvBench			
		Acc_1	Acc_2	Acc_3	Avg	Acc_1	Acc_2	Acc_3	Avg	S_1	S_2	S_3	Avg
LLaVA-1.5-7b	Base	76.72	77.51	77.30	77.18	61.76	64.07	65.35	63.73	4.33	5.72	5.55	5.20
	Random	66.36	66.94	66.68	66.66	54.60	57.07	59.31	56.99	3.73	3.99	4.08	3.93
	Sample	67.11	67.52	67.63	67.42	55.06	57.89	59.74	57.56	3.64	4.85	3.81	4.10
	FastV	45.98	48.56	49.65	48.06	40.98	46.71	49.30	45.66	1.56	1.39	3.12	2.02
	PruMerge	69.03	69.93	69.73	69.56	55.26	57.23	60.13	57.54	4.51	3.47	3.47	3.82
	Ours	70.27	70.31	71.36	**70.65**	55.95	58.71	60.64	**58.43**	4.33	3.99	4.16	**4.16**
LLaVA-1.5-13b	Base	78.35	79.47	78.92	78.91	62.47	65.21	67.22	64.97	4.33	7.11	5.72	5.72
	Random	67.26	68.05	67.63	67.65	54.63	57.87	60.23	57.58	3.56	5.55	4.42	4.51
	Sample	68.12	68.82	68.47	68.47	55.41	58.53	60.26	58.07	4.16	5.03	4.33	4.51
	FastV	55.36	56.80	57.08	56.41	49.08	53.34	56.14	52.85	2.19	3.47	4.20	3.29
	PruMerge	70.18	71.16	70.70	70.68	55.70	57.94	60.70	58.11	4.51	3.99	5.55	4.68
	Ours	72.70	73.24	72.88	**72.94**	57.02	59.26	62.16	**59.48**	4.68	4.51	6.41	**5.20**
LLaVA-NeXT-7b	Base	80.20	80.86	80.71	80.59	63.83	66.68	67.94	66.15	7.95	11.46	7.58	9.00
	Random	70.65	72.26	72.44	71.78	58.60	61.04	63.00	60.88	5.81	6.59	4.42	5.61
	Sample	70.88	72.32	72.35	71.85	58.46	61.39	63.24	61.03	3.81	7.28	5.72	5.60
	FastV	57.09	59.00	59.27	58.45	46.42	50.55	53.95	50.31	0.00	1.85	1.85	1.23
	Ours	73.83	75.24	75.18	**75.18**	59.43	63.49	65.19	**62.70**	4.16	8.67	9.01	**7.28**
LLaVA-NeXT-13b	Base	81.02	82.32	81.64	81.66	65.45	67.32	69.12	67.30	12.48	13.17	7.97	11.21
	Random	71.86	73.44	73.22	72.84	59.61	61.86	63.43	61.63	7.20	9.19	6.33	7.57
	Sample	71.97	73.84	73.43	73.08	59.69	62.28	63.88	61.95	6.07	10.92	6.07	7.69
	FastV	57.07	59.09	59.14	58.43	47.23	51.34	53.36	50.64	5.17	5.17	1.72	4.02
	Ours	74.62	75.73	75.42	**75.26**	60.78	63.41	65.16	**63.12**	6.93	11.27	6.76	**8.32**
XComposer-2.5-7b	Base	78.80	81.12	81.24	80.39	60.21	63.38	65.01	62.87	12.48	7.00	7.97	9.15
	Random	67.88	70.20	70.49	69.52	52.52	57.01	60.67	56.73	11.35	9.88	7.28	9.50
	Sample	68.46	70.92	70.78	70.05	52.94	57.20	59.69	56.61	12.13	9.53	7.63	9.76
	FastV	72.65	75.02	75.02	74.23	54.84	57.33	58.83	57.00	4.17	4.17	0.00	2.78
	Ours	73.91	76.68	76.70	**75.76**	55.99	58.49	61.55	**58.68**	12.31	9.71	7.63	**9.88**

6. Experiments

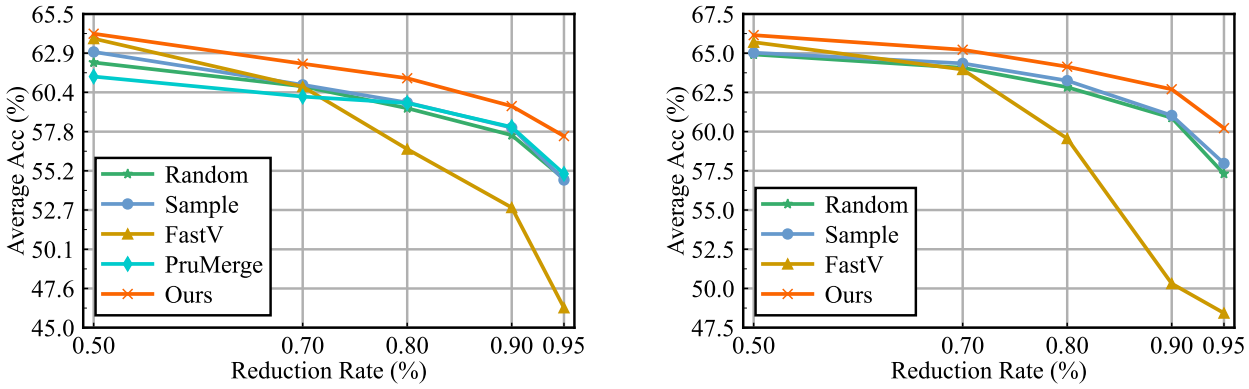
6.1. Implementation

Datasets. We evaluate our method on three MT-VQA benchmarks: MT-VQA-v2, MT-GQA, and ConvBench[2]. MT-VQA-v2 is constructed based on the validation set of VQA-v2 [3, 68] with 25k three-turn image-dialogue pairs. Similarly, MT-GQA is constructed from the testdev-balanced set of GQA [28] with 4061 three-turn dialogues. ConvBench [38] is a native multi-turn conversation evaluation benchmark with 577 conversations that adopts a three-level multimodal capability hierarchy. Instead of training on the entire dataset, which is time-consuming, we only train MetaCompress on a small subset (about 20k items) drawn from the training-balanced split of MT-GQA and the training set of MT-VQA-v2. We utilize the pre-trained weights on MT-VQA-v2 to evaluate on ConvBench.

[2]As ConvBench relies on GPT-3.5-turbo's commercial API for evaluation, we replace it with the recently released open-source LVLM, Llama-3.1-8B-Instruct [18].

LVLMs. To evaluate the generalizability of our method, we choose five different LVLMs: LLaVA-1.5-7b/13b [35], LLaVA-NeXT-7b/13b [37], and InternLM-XComposer-2.5-7b [70]. Of these models, LLaVA-1.5 employs a single-scale vision tower with a fixed visual sequence length, while the others adopt multi-scale perception, resulting in variable visual sequence lengths, which brings further challenges to the token reduction method.

Training Details and Selection of Hyperparameters. We implement our method with PyTorch [44] and optimize the proposed MetaCompress with SGD [47] with a learning rate of 10^{-3}. Gradient clipping is adopted with a maximum value of 10^{-2}. We train all the settings for 2 epochs with a batch size of 36 on four commercial NVIDIA RTX A6000 GPUs. The training of LLaVA-NeXT-7B with a 90% reduction rate takes approximately 30 GPU hours, which corresponds to about only 9 hours on a 4-GPU machine. We initialize $W_q = W_k$ and drawn from Gaussian distribution $\mathcal{N}(0, \frac{1}{\sqrt{d_c}}^2)$; ω is set to all ones; $\alpha_{\text{entropy}} = \alpha_{\text{collapse}} = 1$ as the default setting.



(a) LLaVA-1.5-13b

(b) LLaVA-NeXT-7b

Figure 3. Comparison of average accuracy on MT-GQA with reduction rate from 50% to 95%.

6.2. Comparison Results

We evaluate the proposed MetaCompress and comparison baselines with the following settings:

- **Base:** The base LVLM evaluated directly on the MT-VQA benchmarks without token reduction.
- **Random:** Randomly prune visual tokens before the first layer of the LLM decoder. We report the average performance over 3 random seeds.
- **Sample:** Perform equidistant down-sampling on the visual sequence before the LLM decoder.
- **FastV:** Our implementation of FastV [17] for multi-scale vision tower. The guidance attention weights are extracted from the first layer of the LLM decoder and visual tokens are pruned at the second layer.
- **PruMerge:** Perform LLaVA-PruMerge [35] only for LLaVA-1.5, as it is not compatible with the multi-scale visual tower.
- **Ours:** Perform our proposed MetaCompress before the LLM decoder.

The comparison results with reduction rate 90% and 70% are shown in Table 1 and Table 4, where we compare the accuracy of each turn conversation and the overall accuracy on three MT-VQA benchmarks. *It is noticeable that the proposed MetaCompress consistently outperforms the baseline methods.* While not trained on ConvBench, our method still surpasses the baseline methods by a large margin, demonstrating the transferability of MetaCompress. For LLaVA-1.5, experimental results show that LLaVA-PruMerge which is designed specifically for it performs slightly better than Sample, but still lags behind our approach. On the other hand, FastV performs significantly worse than both the Sample and even the Random methods. This further supports our findings in Section 4, where we have revealed that using attention as guidance for compression results in a loss of critical tokens. Although FastV shows some improvement for XComposer-2.5-7b, it still performs poorly on ConvBench.

Table 2. Efficiency comparison of different token reduction methods. The time to first token (TTFT, ms), end-to-end generation time (E2ET, ms), GPU memory usage (Mem. GB), and TFLOPs are reported on MT-GQA dataset with a reduction rate of 90%.

Model	Setting	TTFT	E2ET	Mem.	TFLOPs
LLaVA-1.5-7b	Base	232 (± 5.8)	676 (± 8.0)	26.9	71.4
	Random	98.2 (± 5.7)	487 (± 4.6)	26.2	13.3
	Sample	96.9 (± 5.8)	482 (± 4.9)	26.2	13.3
	FastV	102 (± 4.52)	528 (± 6.0)	26.3	13.5
	PruMerge	107 (± 5.23)	509 (± 4.5)	26.2	13.3
	Ours	97.8 (± 5.41)	480 (± 5.1)	26.1	13.3
LLaVA-NeXT-7b	Base	484 (± 4.7)	830 (± 13.5)	16.7	95.3
	Random	174 (± 3.4)	481 (± 4.5)	14.8	12.7
	Sample	176 (± 3.2)	484 (± 5.3)	14.8	12.7
	FastV	219 (± 5.0)	529 (± 5.33)	19.2	12.9
	Ours	174 (± 6.1)	501 (± 4.8)	14.9	12.7

Figure 3 illustrates the performance curve of different methods with varying reduction rates from 50% to 95%. It is clear that our method consistently outperforms the baselines across different reduction rates, while FastV performers better for low reduction rate and LLaVA-PruMerge performers better for high reduction rate.

6.3. Efficiency Results

Table 2 compares the inference efficiency of different token reduction methods. As we can observe that our method achieves compatible efficiency with the 'Sample' setting, which is the most efficient baseline thanks to the explicit low-ranking mechanism as described in Equation (9).

6.4. Transfer Results

Beyond the transfer results on ConvBench in Table 1, we further evaluate the transfer capability of MetaCompress through comprehensive cross-dataset validation. Specifically, we perform transfer learning experiments between MT-GQA and MT-VQA-V2, with the results summarized in Table 7. This table reports the average accuracy under a

Table 3. Ablation study of training MetaCompress for LLaVA-NeXT-7b using different loss terms on MT-GQA. Gradient clipping is only applied for the '$\mathcal{L}_{\text{collapse}}$ + *Grad Clip*' setting.

Settings				MT-GQA
$\mathcal{L}_{\text{pred}}$	$\mathcal{L}_{\text{entropy}}$	$\mathcal{L}_{\text{collapse}}$	$\mathcal{L}_{\text{collapse}}$ + *Grad Clip*	*Avg*
✓	✗	✗	✗	61.98
✓	✓	✗	✗	62.42
✓	✗	✓	✗	56.34
✓	✗	✗	✓	62.13
✓	✓	✗	✓	62.70

90% token reduction rate for both directions of transfer, from MT-GQA to MT-VQA-V2 and vice versa. These results indicate that MetaCompress is not heavily dependent on a specific training dataset, demonstrating robust generalization. We also conduct transfer experiments on the video question answering task, as reported in Table 8 of Section 10.4.

6.5. Ablation Study

As delineated in Section 5.3, we utilize three optimization objectives to train the proposed method. To investigate the effectiveness of each objective, we conduct an ablation study by removing one of the objectives at a time. The results in Table 3 (with additional results for various LVLMs in Table 6) demonstrate that each objective contributes positively to the overall performance. In particular, training utilizing the $\mathcal{L}_{\text{collapse}}$ alone leads to divergence because of the relatively high penalty on the collapse objective, especially when the reduction rate is small (less than 70%). To tackle this, we introduce gradient clipping to stabilize the training process.

Besides, we also investigate the sensitivity of the hyperparameters α_{entropy} and α_{collapse} when training MetaCompress. Figure 4 shows the performance curves for different weight settings, demonstrating that the performance remains relatively stable (within a 0.5 percentage point variation).

6.6. Visualization

Figure 5 visualizes the attention distribution for LLaVA-NeXT-7b, similar to Figure 1. As directly computing the attention to [CLS] token is not feasible for multi-scale vision towers, we compute FastV's style image token importance instead. Nevertheless, we observe that only a small number of tokens with high attention are retained, which is consistent with the conclusion in Section 4 and further demonstrates that using token attention to guide reduction is suboptimal.

7. Conclusion and Outlook

This paper proposes a novel token reduction approach for multi-turn VQA scenarios. To this end, we first unify token pruning and merging under the framework of compression projection to visual sequences and explore the optimal compression mapping for a single image. Preliminary results reveal that existing methods guided by attention are suboptimal, as a large number of retained tokens do not correspond to the highest attention scores. This motivates us to further explore the construction of an optimal compression mapping for the entire dataset. To achieve this, we propose MetaCompress, a meta generator conditioned solely on the visual sequence, and optimized in a data-driven manner. Extensive experiments demonstrate the efficiency and effectiveness of our method. In future work, we will explore the token reduction strategy for all LLM layers without a hand-crafted design, and investigate the transferability of our method to more challenging tasks, such as video understanding.

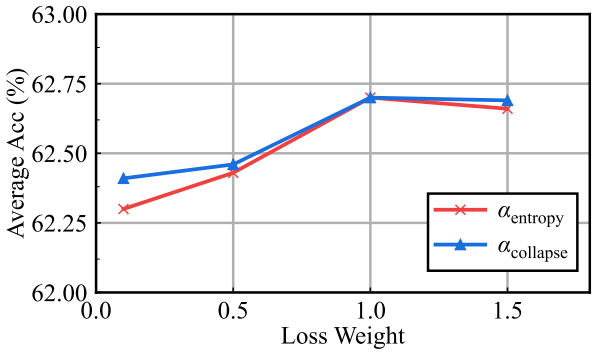


Figure 4. Sensitivity analysis in training MetaCompress for LLaVA-NeXT-7b with different weights α_{entropy} and α_{collapse} on MT-GQA.

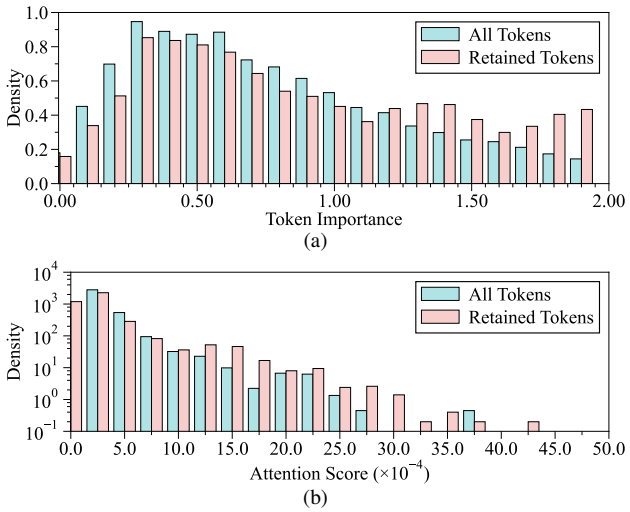


Figure 5. (a) Token importance distribution. (b) Attention distribution over prompt tokens. Image tokens are extracted from the last layer of the vision tower of LLaVA-NeXT-7b on VQA-v2 dataset.

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